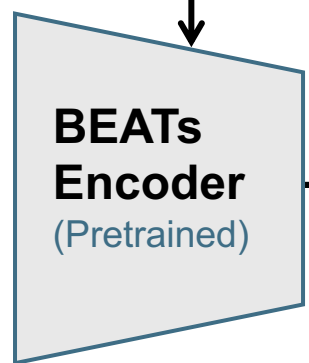


Respiratory Audio



H_i
 $T_i \times D$



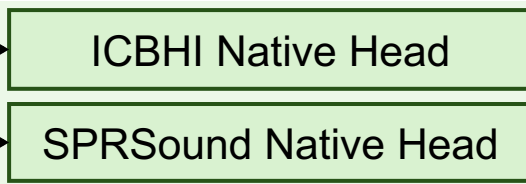
h_i



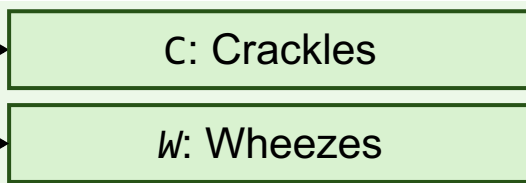
$\mathbb{R}^D$

Z_i

Task Heads



Shared C-W Heads



Training Supervision



HF train* (Optional)

$$\mathcal{L} = \lambda_p \mathcal{L}_{PCSL} + \lambda_g \mathcal{L}_{GPAL} + \mathcal{L}_{cls} + \lambda_{HF} \mathcal{L}_{HF}$$

Patient Contrastive

Patient Alignment

Native + C-W Classification

HF Supervision (Optional)

Task-Specific Outputs

ICBHI*:
Normal · Crackle · Wheeze · Both

SPRSound*:
Normal · Adventitious (Abnormal)

KAUH#:
Normal · Abnormal

HF test#:
Crackle · Wheeze · Both

*: Training + Inference

#: Inference Only